

Our billions of nonsense AI conversations are consuming so much energy

Posted by u/soonerfan360 • 0 points • 14 comments

Just a simple 4-5 question convo about how open AI is going to keep up with Anthropic, 1500 tokens...Enough to power LED bulb for over 1 hour!

Comments

u/[deleted] • 1 points • 2025-05-16 03:07 UTC

[removed]

u/Deciheximal144 • 1 points • 2025-05-16 02:44 UTC

So can I go down to the OpenAI data center to enjoy some of their freshly toasted pop tarts?

u/mop_bucket_bingo • 2 points • 2025-05-16 02:43 UTC

There are people who want you to be upset all the time about everything. This is, unsurprisingly, one of the things they want you to be upset about.

u/epicchad29 • 1 points • 2025-05-16 02:42 UTC

You can't even measure power consumption per token as the cost is exponential for longer conversations. The amount of energy necessary to compute a token is some constant times t^k where t is the number of previous tokens in the sequence, and k is some constant. (I think its roughly 2.3, but OpenAI may have some slight performance improvements in house).

u/DueCommunication9248 • 1 points • 2025-05-16 02:36 UTC

Wait still we get 10GW data centers

u/grass_is_scary • 1 points • 2025-05-16 09:57 UTC

why not 1000GW data centers

u/oldboi777 • 3 points • 2025-05-16 02:23 UTC

I just read it takes less power than running video games

u/Rhawk187 • 3 points • 2025-05-16 02:19 UTC

I've toasted so many pop tarts with this thing.

u/ohwut • 11 points • 2025-05-16 02:19 UTC

As always. Don't believe a single word ChatGPT says.

First it says 1,000 tokens is 5 to 10 cents in electricity. That would mean it's using ~750Wh of electricity.

The. It says your 1,500 token conversation is 30-150Wh. So what is it?

In reality 1,000 tokens is more like 4-5Wh or \$0.00065 in electricity.

u/HidingInPlainSite404 • 1 points • 2025-05-16 02:54 UTC

Put your comment in a chatbot and ask what was wrong in your logic, if you can't spot it.

u/ohwut • 3 points • 2025-05-16 03:05 UTC

Sure:

"Conclusion

- You're right to call out absurd overestimates like 750 Wh.
- Your energy numbers are reasonable for GPT-4-level inference.
- But total per-token cost includes much more than just energy, and any implied equivalence would be misleading.

Let me know if you want sources, deeper breakdowns, or cost curves by model size."

u/HidingInPlainSite404 • 0 points • 2025-05-16 03:37 UTC

****Commenter's Claim:**** "First it says 1,000 tokens is 5 to 10 cents in electricity. That would mean it's using $\sim 750\text{Wh}$ of electricity."

*** **The commenter's mistake:**** The commenter is deriving an energy usage ($\approx 750\text{ Wh}$) from ChatGPT's *cost estimate* (5 to 10 cents) by assuming a typical electricity price (around 6.7 to 13.3 cents/kWh would yield that 750 Wh).

*** **Why this is misleading:**** The ChatGPT post *also* provided its own figures for energy consumption. From Image 2 (the "TL;DR" part), a single token is ≈ 0.00002 to 0.0001 kWh .

*** Therefore, for 1,000 tokens, ChatGPT's stated energy use is:** $1000\text{ tokens} \times (0.00002\text{ to }0.0001\text{ kWh/token}) = 0.02\text{ to }0.1\text{ kWh}$.

*** Converting this to Watt-hours (Wh):** $0.02\text{ kWh} = 20\text{ Wh}$, and $0.1\text{ kWh} = 100\text{ Wh}$.

*** **Conclusion:**** So, ChatGPT's own figures state that 1,000 tokens use **20 to 100 Wh**, not the $\sim 750\text{Wh}$ the commenter calculated based on cost alone. The commenter ignored ChatGPT's explicit energy data and created a number that doesn't align with what ChatGPT actually said about its energy usage.

u/HidingInPlainSite404 • 0 points • 2025-05-16 03:38 UTC

Want me to keep going?:

****Commenter's Claim:**** "In reality 1,000 tokens is more like 4-5Wh or \$0.00065 in electricity."

*** This is the commenter's own figure.** 4 to 5 Wh for 1000 tokens ($0.004\text{ to }0.005\text{ Wh/token}$) is significantly lower than ChatGPT's estimate of 20 to 100 Wh for 1000 tokens.

*** If 4 to 5 Wh ($0.004\text{ to }0.005\text{ kWh}$) costs \$0.00065 (which is 0.065 cents), the implied electricity price is $0.065\text{ cents} / 0.004\text{ kWh} \approx 16.25\text{ cents/kWh}$ or $0.065\text{ cents} / 0.005\text{ kWh} = 13\text{ cents/kWh}$. These are plausible electricity prices.**

*** **Is the commenter's "reality" correct?**** Energy consumption for AI tokens is complex and varies widely based on the specific model, hardware, and data center efficiency.

*** ChatGPT's stated range ($0.02\text{ to }0.1\text{ Wh/token}$) is actually in line with some public estimates for large language models.**

*** The commenter's estimate ($0.004\text{ to }0.005\text{ Wh/token}$) is on the very low side compared to many public discussions about models like GPT-3.5/4, but could be possible for smaller, highly optimized models or different types of AI tasks. They present this as "reality" without providing a source or context.**

u/ohwut • 3 points • 2025-05-16 03:58 UTC

Enjoy your hallucinations.

Best of luck with your life.